

Introduction to Computer Programming

2026S_project_guide1



Outline

- **Project Structure**
- **Window Size**
- **Board Size**
- **Login Window**



Project Structure

resource

0.png

1.png

图片、音频等资源

src

app

Main

主程序，程序入口

model

Cell

GameBoard

Line

Position

Rectangle

- Cell 格子类
- GameBoard 棋盘类
- Line 连线类
- Position 坐标位置类
- Rectangle 格子边框（选中框）

ui

BoardPanel

ControlPanel

GameFrame

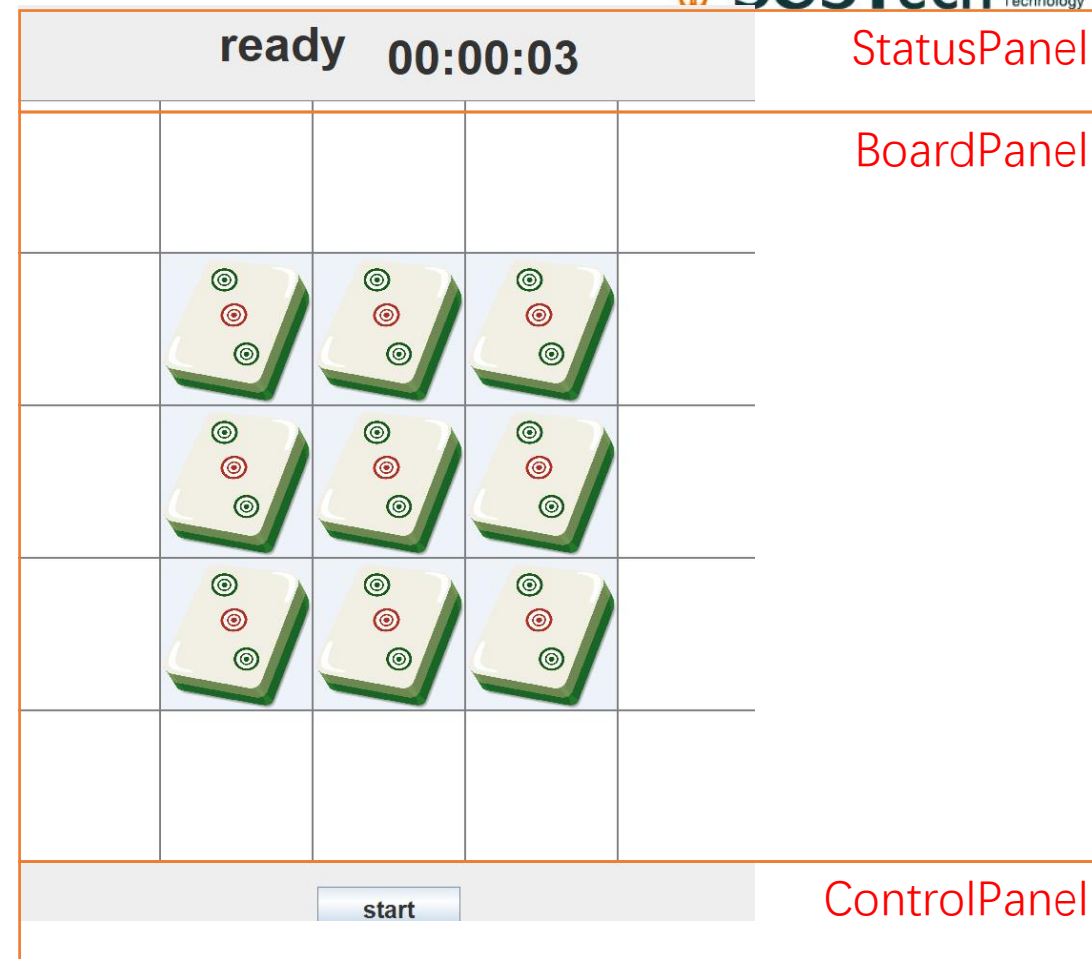
StatusPanel

- BoardPanel 棋盘面板
- ControlPanel 控制面板
- GameFrame 游戏主窗口
- StatusPanel 状态面板

utils

Utils

- Utils 判断两个Cell是否可以合法连线的一些静态方法



Window Size

src\app\Main.java

```
public class Main {  
    public static void main(String[] args) {  
        SwingUtilities.invokeLater(() -> {  
            GameFrame frame = new GameFrame("连连看", 800, 1000);  
            frame.repaint();  
        });  
    }  
}
```

窗口标题

窗口大小



Board Size

src\ui\GameFrame.java

```
public GameFrame(String title, int width, int height) {  
    super(title);  
    this.setResizable(false);
```

```
    int size = 3;  
    Cell[][] board = new Cell[size + 2][size + 2];  
    for (int i = 0; i < size + 2; i++) {  
        for (int j = 0; j < size + 2; j++) {  
            if (i == 0 || i == size + 1 || j == 0 || j == size + 1) {  
                board[i][j] = new Cell(new Position(i, j), true, 0);  
            }  
        }  
    }  
    for (int i = 1; i <= size; i++) {  
        for (int j = 1; j <= size; j++) {  
            board[i][j] = new Cell(new Position(i, j), false, 1);  
        }  
    }  
}
```

网格绘制

边框 空格子

棋盘大小

有棋子的格子

```
BoardPanel boardPanel = new BoardPanel(new GameBoard(5, 5, board), 0, 100, 800, 800);  
this.title = title;
```

棋盘面板大小



JLabel & JButton

```
public class StatusPanel extends JPanel
{
    JLabel statusLabel;
    ....
    this.add(statusLabel);
}
```

添加文本框

```
public void setStatus(String text) {
    statusLabel.setText(text);
    .....
    repaint();
}
```

更新文本框文本

src\ui\StatusPanel.java

```
this.statusPanel_ = new StatusPanel(0, 0, 800, 100);
```

```
this.controlPanel = new ControlPanel(statusPanel, 0, 900, 800, 100);
```

statusPanel对象作为参数传入
controlPanel用于更新状态栏

```
public ControlPanel(StatusPanel statusPanel, int offSetX, int
offSetY, int width, int height) {
```

src\ui\ControlPanel.java

.....

```
this.startButton = new JButton("start");
```

创建按钮

```
this.statusPanel = statusPanel;
```

.....

```
this.add(startButton);
```

添加按钮到当前Panel

```
this.startButton.addActionListener(e -> {
```

```
    statusPanel.setStatus("RUN");
```

添加按钮点击事件

```
});
```

```
}
```

src\ui\GameFrame.java

Login window

```
JFrame loginFrame=new JFrame("登录");
loginFrame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
loginFrame.setSize(400,200);
loginFrame.setLayout(null);
JTextField textField=new JTextField();
textField.setSize(100,50);
textField.setLocation(50,50);
JButton login = new JButton("按钮");
login.setLocation(50,100);
login.setSize(100, 50);
login.addActionListener(e -> {
    //添加事件
});
loginFrame.add(login);
loginFrame.add(textField);
loginFrame.setVisible(true);
```

放这里

放这里

src\app\Main.java

```
public class Main {
    public static void main(String[] args) {
        SwingUtilities.invokeLater(() -> {
            GameFrame frame = new GameFrame("连连看", 800, 1000);
            frame.repaint();
        });
    }
}
```

- JTextField 输入框
- JLabel 文本标签
- JButton 按钮

在按钮添加事件处
添加读写文件的代码
可以实现登录注册等功能



THANK YOU

